GLP-1 Utilization & Health Needs Across U.S. States

IS 507 · Project Proposal **(Group1)**

Motivation

GLP-1 medications have become increasingly popular for diabetes and weight management in the United States. However, their utilization may vary across states, even among populations with similar health needs. **Medicaid** provides healthcare coverage to many low-income individuals, making it important to examine how Medicaid GLP-1 prescription patterns relate to state-level obesity and diabetes prevalence.

> *"I am a data scientist at a public health organization tasked with examining whether GLP-1 prescription patterns align with healthcare needs across U.S. states. The available datasets include Medicaid GLP-1 prescription records, Medicaid enrollment data, and state-level obesity and diabetes prevalence data."*

Goal

"Are U.S. states with Medicaid GLP-1 prescription rates below the levels predicted by their obesity and diabetes prevalence catching up during 2025?"

We aim to identify states with Medicaid GLP-1 prescription rates below model-predicted levels and examine whether these gaps narrowed from Q1 to Q4 of 2025. In this study, **"catching up"** is defined as a reduction in the gap between actual and predicted prescription rates. These model-based gaps represent statistical deviations from predicted utilization, not direct evidence of unmet medical needs or barriers to access.

- **X (Input Features):** State-level obesity prevalence, diabetes prevalence, and quarter (Q1–Q4)
- **Y (Target Variable):** Medicaid GLP-1 prescription rate, calculated as the number of GLP-1 prescriptions divided by average Medicaid enrollment for each state and quarter

$$Y_{i,t} = \frac{GLP{-}1\ Prescriptions_{i,t}}{Average\ Medicaid\ Enrollment_{i,t}}$$

In this study, Medicaid GLP-1 utilization is measured as the number of GLP-1 prescriptions per Medicaid enrollee per quarter. This measure reflects **prescription activity** rather than the proportion of individuals using GLP-1 medications.

- **Loss Function: Mean Squared Error (MSE)**, which measures the average squared difference between predicted and actual GLP-1 prescription rates

$$MSE = \frac{1}{n}\sum_{j=1}^{n}(Y_j - \hat{Y}_j)^2$$

Potential impact

This analysis could help public health organizations identify states where Medicaid GLP-1 prescription rates are relatively low compared with model-predicted levels based on obesity and diabetes prevalence. Examining changes in these gaps throughout 2025 could reveal which states are moving closer to predicted utilization levels and which continue to show persistent shortfalls.

These findings may help policymakers and public health organizations identify states that warrant further investigation into **potential disparities in GLP-1 prescribing patterns** and **access to treatment**.

Related work

Do et al. (2026) examined state-level GLP-1 utilization using 2023 prescription claims data and found weak correlations between disease prevalence and GLP-1 utilization among Medicaid populations. Their findings suggest that obesity and diabetes prevalence alone may not fully explain differences in GLP-1 utilization across states.

Building on this research, we extend the analysis to 2025 by examining quarterly changes in state-level utilization gaps. Using **multiple linear regression**, we will identify states with lower-than-predicted utilization and assess whether these gaps narrowed from Q1 to Q4.

Data

1. **Medicaid Drug Utilization — CMS**
   The Medicaid Drug Utilization dataset contains prescription drug utilization data reported by states.
   - **Dataset Dimension:** 5,308,547 rows × 15 columns
   - **Unit of Observation:** One National Drug Code (NDC)-level drug utilization record for a specific state and quarter

- **Key Variables (Types):** State Code (str), NDC (str), Year (int), Quarter (int), Product Name (str), Number of Prescriptions (int), and Suppression Used (bool)
- **Purpose:** Identify GLP-1 medications in 2025 and aggregate prescription counts by state and quarter

2. **2025 Behavioral Risk Factor Surveillance System (BRFSS) — CDC**
   The 2025 BRFSS dataset contains health conditions and demographic information collected through a state-based survey.
   - **Dataset Dimension:** 356,158 rows × 283 columns
   - **Unit of Observation:** One individual survey respondent
   - **Key Variables (Types):** State Identifier (_STATE), BMI Category (_BMI5CAT), Diabetes Status (DIABETE4), Survey Weight (_LLCPWT), and Primary Health Insurance (PRIMINS2) – all stored as float; the first three represent categorical information
   - **Purpose:** Calculate weighted state-level obesity and diabetes prevalence and assess the feasibility of estimating these rates specifically among respondents who report Medicaid as their primary health insurance (PRIMINS2 = 5).

3. **Medicaid Enrollment — CMS**
   The Medicaid Enrollment dataset contains monthly Medicaid enrollment information reported by states.
   - **Dataset Dimension:** 11,118 rows × 44 columns
   - **Unit of Observation:** One state-level enrollment record for a specific reporting period
   - **Key Variables (Types):** State Abbreviation (str), State Name (str), Reporting Period (int), Preliminary or Updated (str), and Total Medicaid Enrollment (float)
   - **Purpose:** Obtain Medicaid enrollment data for 2025 and align monthly enrollment figures with quarterly prescription counts

Data Cleaning and Preparation

1. **Variable Selection:** Retain the relevant variables from each dataset and standardize state identifiers and reporting periods.
2. **GLP-1 Identification:** Identify GLP-1 medications using product names listed in the FDA drug safety communication and corresponding NDC information.
3. **Missing and Suppressed Data:** Examine missing values and suppressed prescription records to assess their potential impact on the analysis.
4. **Data Aggregation and Integration:** Aggregate prescription counts and enrollment by state-quarter, estimate survey-weighted health prevalence using

_LLCPWT, assess Medicaid-specific BRFSS estimates using PRIMINS2, and merge the datasets.

Potential Challenges

1. **Suppressed and Missing Prescription Data:** Suppressed prescription counts cannot be treated as zero and may affect utilization estimates.
2. **Missing and Coded Responses in BRFSS:** Unknown, refused, and missing responses must be handled before estimating obesity and diabetes prevalence.
3. **Survey-Weighted Health Estimates:** BRFSS requires survey-weighted estimation to account for its complex sampling design.
4. **Enrollment Records and Data Alignment:** Preliminary and updated enrollment records must be reconciled, and datasets aligned at the state-quarter level.
5. **Temporal Limitations of BRFSS:** Annual prevalence estimates remain fixed across quarters. Although interview dates allow quarterly grouping, uneven state-quarter sample sizes may limit the reliability of quarterly estimates.
6. **Population Differences:** BRFSS covers adults, while Medicaid enrollment includes all ages. Small state-level samples may limit Medicaid-specific estimates using PRIMINS2.
7. **Prescription-Based Utilization Measure:** Prescription counts include repeat prescriptions and cannot be interpreted as the proportion of beneficiaries using GLP-1 medications.

Plan

1. Multiple Linear Regression

We will use **multiple linear regression** to predict Medicaid GLP-1 prescription rates based on state-level obesity prevalence, diabetes prevalence, and quarter (treated as a categorical variable).

We will assess Medicaid-specific BRFSS estimates using sample sizes and estimate precision. If unreliable, we will use general adult estimates and acknowledge the population mismatch.

2. Gap and Trend Analysis

We will calculate the difference between actual and model-predicted GLP-1 prescription rates for each state and quarter:

$$Gap_{i,t} = ActualRate_{i,t} - PredictedRate_{i,t}$$

A negative gap indicates lower-than-predicted utilization. We will track states with negative Q1 gaps through Q4, comparing changes in actual and predicted rates to assess their respective contributions to gap reduction. We will also compare trends with other states and consider regression to the mean.

3. Model Training and Evaluation

We will use **state-grouped cross-validation**, keeping all quarters of each state in the same fold to prevent state-level leakage. Out-of-fold predictions will be used to calculate gaps consistently across Q1–Q4, while evaluating performance on unseen states.

Performance will be evaluated using **Mean Squared Error (MSE)** against the baseline.

If regression fails to outperform the baseline, model-based gaps will not be interpreted as reliable indicators of health-related utilization differences.

4. Baseline

We will use the **quarterly average Medicaid GLP-1 prescription rate across training states** as our baseline, predicting the same rate for all states within each quarter without considering obesity or diabetes prevalence.

Both models will be evaluated using identical state-grouped cross-validation folds, and their MSE values will be compared to assess whether incorporating health indicators improves prediction accuracy.

References

Centers for Medicare & Medicaid Services. (n.d.). *State drug utilization data*. https://data.medicaid.gov/dataset/158a1baa-5506-400a-8ec3-97756f0b0536

Centers for Disease Control and Prevention. (n.d.). *2025 BRFSS survey data and documentation*. https://www.cdc.gov/brfss/annual_data/annual_2025.html

Centers for Medicare & Medicaid Services. (n.d.). *Medicaid enrollment data*. https://data.medicaid.gov/dataset/6165f45b-ca93-5bb5-9d06-db29c692a360

Do, S., Christensen, K., Inneh, A., Peasah, S., & Good, C. (2026). State-level variation in the use of glucagon-like peptide-1 receptor agonists for weight loss and diabetes: A real-world analysis. *Journal of Managed Care & Specialty Pharmacy, 32*(5), 625–630. https://doi.org/10.18553/jmcp.2026.32.5.625

U.S. Food and Drug Administration. (n.d.). *FDA drug product information*. https://www.fda.gov/drugs/drug-safety-communications/update-fdas-ongoing-evaluation-reports-suicidal-thoughts-or-actions-patients-taking-certain-type